

A short tutorial on random number generation

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```
int getRandomNumber()  
{  
    return 4; // chosen by fair dice roll.  
              // guaranteed to be random.  
}
```

<https://xkcd.com/221/>

U[0,1]-distributed random numbers

- A good uniform random generator on the interval $[0, 1]$ is a major component of any good random generator library.
- Draws from other distributions are usually obtained by adequately transforming a uniformly distributed sample.

Define a **transition function** $f : \mathcal{S} \rightarrow \mathcal{S}$, where $\mathcal{S}$ is the **state space**. The cardinality of $\mathcal{S}$ is assumed to be finite.

The initial state is denoted by s_0 , and we will write

$$s_n = f(s_{n-1}).$$

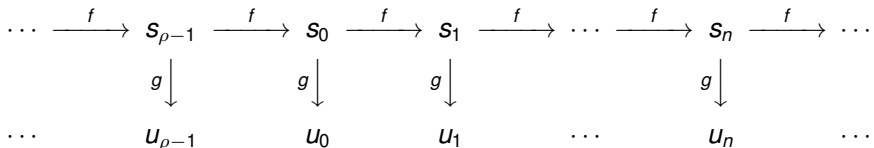
We will furthermore assume that f is periodic for all n greater of equal to some known τ (often equal to 0), with period ρ :

$$s_{n+\rho} = s_n, \quad \forall n \geq \tau.$$

U[0,1]-distributed random numbers (cont'd)

- Output space: $\mathcal{U}$.
- We assume here that $\mathcal{U} = (0, 1)$.
- Output function $g : \mathcal{S} \rightarrow \mathcal{U}$.

It transforms the state s_n into an output value u_n .



How to choose f and g ?

Goals: large ρ , good uniformity, “random” behavior.

Linear congruential generators (LCGs)

- Introduced by Lehmer (1951).
- State $s = x \in \mathbb{N}_{\geq 0}$.
- Recursive formula:

$$x_i = f(x_{i-1}) = (ax_{i-1} + c) \bmod m,$$

where $\bmod$ is the modulo operator.

Given two numbers, a (the dividend) and n (the divisor),

$$a \bmod n = a - n \left\lfloor \frac{a}{n} \right\rfloor.$$

- If $c = 0$, the generator is often called a multiplicative congruential generator, or “**Lehmer RNG**”

Linear congruential generators: full period?

Full period: m is $c \neq 0$, $m - 1$ otherwise (if $c = 0$, 0 is a fixed point for the recurrence). Consider the case $c \neq 0$.

Theorem (Period)

The LCG has full period iff the following three conditions hold:

- 1. the only positive integer that (exactly) divides both m and c is 1;*
- 2. if q is a prime number that divides m , then q divides $a - 1$;*
- 3. if 4 divides m , then 4 divides $a - 1$.*

“Standard minimal” (Park and Miller, 1988):

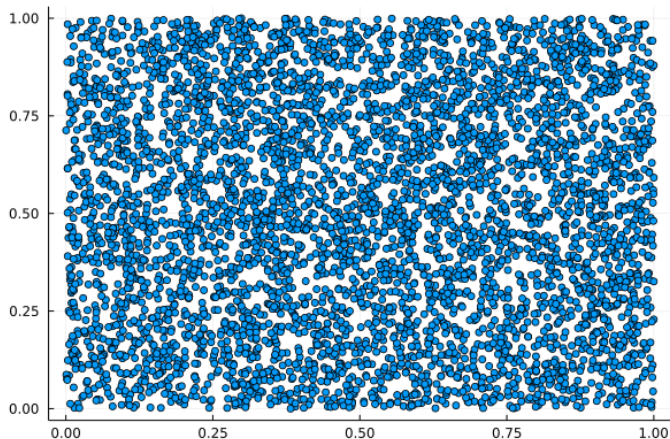
$$x_{n+1} = 16807x_n \bmod 2147483647.$$

Observe that $2147483647 = 2^{31} - 1$; on 32-bit architectures, the largest representable (signed) integer is $2^{31} - 1$.

The Standard Minimal Generator

```
function getlcg(seed::Integer, a::Integer,  
               c::Integer, m::Integer)  
    state = seed  
    am_mil = 1.0/m  
    return function lcgrand()  
        state = mod(a * state + c, m)  
        # produce a number in (0,1)  
        return state*am_mil  
    end  
end  
  
stdmin = getlcg(1234, 16807, 0, 2^31-1)
```

Standard minimal generator: illustration



10000 generated points on the unit square.

Multiple Recursive Generator (MRG)

But we want better! Generalize the LCG:

$$x_n = (a_1 x_{n-1} + \cdots + a_k x_{n-k}) \bmod m, \quad u_n = x_n/m.$$

In practice, take $u_n = (x_n + 1)/(m + 1)$, or $u_n = x_n/(m + 1)$ if $x_n > 0$ and $u_n = m/(m + 1)$ otherwise, but the structure remains the same, and is easier when studying theoretical properties.

State at step n :

$$s_n = (x_{n-k+1}, \dots, x_n)^T.$$

State space: $\mathcal{Z}_m^k$, of cardinality m^k .

The maximal period is $\rho = m^k - 1$.

Period of MRGs

It can be shown that for $k > 1$, it is sufficient to have at least two non-zero coefficients, including a_k , in order to get the maximal period.

The cheapest recurrence has therefore the form

$$x_n = (a_r x_{n-r} + a_k x_{n-k}) \mod m.$$

But how to choose a_r and a_k ?

It is possible to study theoretical properties of MRG's, and exclude directly some generators that have known strong deficiencies.

Choosing a good MRG

Example: Lagged-Fibonacci

$$x_n = (\pm x_{n-r} \pm x_{n-k}) \mod m.$$

It can be shown the vectors $(u_n, u_{n+k-r}, u_{n+k})$ are all contained in two planes! We therefore know without additional tests that the numbers cannot be considered as random.

In practice, we can impose various conditions on the coefficients, and compute theoretically appealing generators by maximizing some quality measure. This maximization is numerically expensive.

Combined MRG's

Consider two (or more) MRG's working in parallel:

$$\begin{aligned}x_{1,n} &= (a_{1,1}x_{1,n-1} + \cdots + a_{1,k}x_{1,n-k}) \bmod m_1, \\x_{2,n} &= (a_{2,1}x_{2,n-1} + \cdots + a_{2,k}x_{2,n-k}) \bmod m_2.\end{aligned}$$

We define the two combinations

$$\begin{aligned}z_n &:= (x_{1,n} - x_{2,n}) \bmod m_1; & u_n &:= z_n/m_1; \\w_n &:= (x_{1,n}/m_1 - x_{2,n}/m_2) \bmod 1.\end{aligned}$$

The sequence $\{w_n, n \geq 0\}$ is the output of another MRG, of modulus $m = m_1 m_2$, and $\{u_n, n \geq 0\}$ is nearly the same sequence if m_1 and m_2 are close.

We can achieve the period $(m_1^k - 1)(m_2^k - 1)/2$.

MRG32k3a

The following combined MRG was proposed by L'Ecuyer, and is amongst the most popular and efficient known generators. It combines 2 MRG's.

$k = 3$,

$m_1 = 2^{32} - 209$, $a_{11} = 0$, $a_{12} = 1403580$, $a_{13} = -810728$,

$m_2 = 2^{32} - 22853$, $a_{21} = 527612$, $a_{22} = 0$, $a_{23} = -1370589$.

Combination: $z_n = (x_{1,n} - x_{2,n}) \bmod m_1$.

Corresponding MRG: $k = 3$,

$m = m_1 m_2 = 18446645023178547541$,

$a_1 = 18169668471252892557$,

$a_2 = 3186860506199273833$,

$a_3 = 8738613264398222622$.

Period $\rho = (m_1^3 - 1)(m_2^3 - 1)/2 \approx 2^{191}$.

MRG32k3a: basic implementation

```
function rand(rng::MRG32k3a)

p1::Int64 = (a12 * rng.Cg[2] + a13 * rng.Cg[1]) % m1
p1 += p1 < 0 ? m1 : 0

rng.Cg[1] = rng.Cg[2]
rng.Cg[2] = rng.Cg[3]
rng.Cg[3] = p1
p2::Int64 = (a21 * rng.Cg[6] + a23 * rng.Cg[4]) % m2
p2 += p2 < 0 ? m2 : 0

rng.Cg[4] = rng.Cg[5]
rng.Cg[5] = rng.Cg[6]
rng.Cg[6] = p2
u::Float64 = p1 > p2 ? (p1 - p2) * norm :
    (p1 + m1 - p2) * norm
end
```

MRG32k3a: implementations

Note: more efficient implementations exist.

See for instance <https://github.com/vigna/MRG32k3a>.

Random numbers generators on $\mathcal{F}_2$

MRGs work on integers. An alternative is to build generators directly on **bits**, with **linear recurrences in $\mathcal{F}_2$** .

Recall the Galois field $\mathcal{F}_2$: the set $\{0, 1\}$ with addition and multiplication computed modulo 2.

All operations are then **bitwise**: shifts, xors and masks, which makes these generators **numerically very fast**.

Generators on $\mathcal{F}_2$: state and output

We construct two sequences of bit vectors, the **state** and the **output**, with linear recurrences (k -bit state, w -bit output):

$$\begin{aligned}\mathbf{x}_n &= X\mathbf{x}_{n-1} && \text{(state vector, } k \text{ bits),} \\ \mathbf{y}_n &= B\mathbf{x}_n && \text{(output vector, } w \text{ bits),}\end{aligned}$$

where X and B are binary matrices and all arithmetic is in $\mathcal{F}_2$.

Reading a binary fraction. The w output bits are stacked after the radix point:

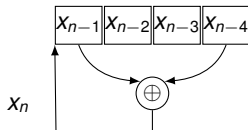
$$u_n = \sum_{j=1}^w y_{n,j-1} 2^{-j} = .y_{n,0} y_{n,1} \cdots y_{n,w-1}.$$

Example: $y_n = (1, 0, 1, 1)$ gives $2^{-1} + 2^{-3} + 2^{-4} = 0.6875 = .1011_2$.

LFSR: a shift register with linear feedback

A **linear feedback shift register** (LFSR) is a k -bit register updated in three steps:

1. read the **tapped bits** — here the positions r and k ;
2. **xor** them: the new bit is $x_n = x_{n-r} \oplus x_{n-k}$ (it is read before the shift);
3. **shift** the register by one position and insert x_n .



In $\mathcal{F}_2$, addition is the **exclusive-or** (xor): $0 \oplus 0 = 0$, $0 \oplus 1 = 1$, $1 \oplus 1 = 0$.

LFSR: a tiny example ($k = 4$, taps $r = 1$ and $r = 4$)

Seed 0001, recurrence $x_n = x_{n-1} \oplus x_{n-4}$:

step	register				new bit
	x_{n-1}	x_{n-2}	x_{n-3}	x_{n-4}	x_n
0	0	0	0	1	—
1	1	0	0	0	1
3	1	1	1	0	1
7	0	1	0	1	0
9	1	1	0	1	1
$\vdots$					$\vdots$

- After 15 steps we are back to the seed 0001: period $2^4 - 1$.
- All $2^4 - 1$ non-zero states have been visited: maximal period.

LFSR: matrix form

The recurrence is a linear recurrence over $\mathcal{F}_2$ on the state vector (we assume $a_k \neq 0$):

$$\mathbf{x}_n = X \mathbf{x}_{n-1}, \quad X = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ a_k & a_{k-1} & \dots & a_1 \end{pmatrix}.$$

- The **last row** encodes the taps (the feedback coefficients), the rest of the matrix is a pure shift.
- X^ν shifts the register by ν bits in one step: this is the spacing ν used when assembling output words.
- For **Tausworthe** generators, $B = I$ (no output transformation).

Tausworthe generators: assembling words

Tausworthe (1965): read the bit stream w bits at a time, spaced ν apart, and interpret them as a binary fraction:

$$u_n = \sum_{\ell=1}^w x_{n\nu+\ell-1} 2^{-\ell} = .x_{n\nu} x_{n\nu+1} \dots x_{n\nu+w-1}.$$

Example (previous LFSR, $w = 3$, $\nu = 1$): the output bit stream is

$$\begin{array}{ccccc} \underbrace{111} & \underbrace{101} & \underbrace{011} & \underbrace{001} & \underbrace{000} \\ \frac{7}{8} & \frac{5}{8} & \frac{3}{8} & \frac{1}{8} & 0 \end{array}$$

so the uniforms are $u = .111, .101, .011, .001, .000, \dots$

Tausworthe generators: maximal period

Maximum period. The output period reaches its maximal value

$$\rho = 2^k - 1$$

iff the characteristic polynomial is **primitive** and the spacing is **coprime** to the period:

$$Q(z) = z^k - a_1 z^{k-1} - \dots - a_{k-1} z - a_k \text{ primitive,}$$

$$\gcd(\nu, 2^k - 1) = 1.$$

In most applications, only two coefficients are non-zero:

$$Q(z) = z^k - a_r z^{k-r} - a_k,$$

which, in $\mathcal{F}_2$, brings us back to the xor recurrence $x_n = x_{n-r} \oplus x_{n-k}$ of the LFSR.

Caveat. Linearity in $\mathcal{F}_2$ is a strong structural flaw: plain LFSRs/Tausworthe are **not good on their own**, and are combined, or replaced by MT19937 and xoshiro (next slide).

Implementations

More generally, we construct a fast implementation by using shifts, xor's, masks,... We can also combine them.

Most popular:

- Mersenne Twister MT19937 (Matsumoto and Nishimura); period of $2^{19937} - 1$
- xoshiro: <https://prng.di.unimi.it/>; by default, Julia uses `xoshiro256++`

But even these recent generators are not without flaws! See e.g.

- It Is High Time We Let Go Of The Mersenne Twister
- Unveiling patterns in xorshift128+ pseudorandom number generators

PCGs

- One of the main opponent of the xorshift family is Melissa O'Neil, who proposes the PCGs generators: <https://www.pcg-random.org/>.
- PCG combines a linear congruential generator and permutation functions.
- However, Vigna (the author of xorshift generators) has a lot of criticisms regarding PCG: <https://pcg.di.unimi.it/pcg.php>.
- In summary, there are no strong evidence that one should favor a PCG over a MRG. According to Vigna, "there is technically no sensible reason to use a PCG generator: those without flaws are not competitive."

Counter-based generators

- In previously covered RNGs, the transition function does most of the transformation and the output function is very simple.
- Opposite for CBRNGs: the state is a counter, increased by 1, and the output function is complex.
- The most popular implementation, Philox, detailed next, is the default generator in TensorFlow.
- They pass demanding test suites such as TestU01's BigCrush, with sufficiently many rounds, and are efficient on both CPUs and GPUs.
- One strength: easy to jump ahead in the state space.

Counter-based generators: a stateless map

A CBRNG is a **stateless map** onto a block of random bits:

$$R = E_K(C),$$

where

- $C = (c_1, \dots, c_N)$ is a **counter**: the position in the sequence;
- $K = (k_1, \dots, k_M)$ is a **key**: it identifies the stream; for Philox, $M = N/2$.

The sequence follows by **incrementing** the counter:

$$C, C + 1, C + 2, \dots$$

Consequence: identical (C, K) always yields identical bits on any architecture — ideal for reproducible parallel Monte Carlo.

Philox: an encrypt-increment CBRNG

Introduced by [Salmon, Moraes, Dror & Shaw](#) (SC 2011), together with Threefry and AES-based designs: the *Random123* library,

<https://github.com/DEShawResearch/random123>.

- **Default variant** (PyTorch, TensorFlow, NumPy `Generator`, JAX, cuRAND):

Philox $4 \times 32 - 10$, $N = 4$, $W = 32$ bits, $r = 10$ rounds,

i.e. a block of $4 \times 32 = 128$ bits per call.

- **Not cryptographic**: the few rounds are meant to pass statistical tests, not to resist cryptanalysis (e.g. the key space is finite: 2^{64} keys).
- Fewer rounds are faster but display measurable correlations.

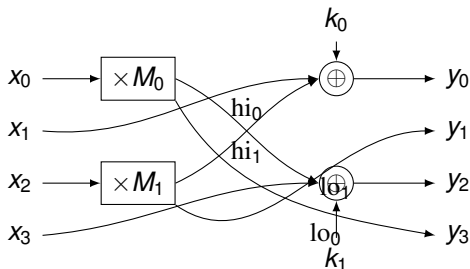
Philox: one round — four elementary operations

Each round is a bijection on (x_0, x_1, x_2, x_3) , given two subkeys (k_0, k_1) . It alternates **arithmetic** (multiplication) and **bitwise** (xor) mixing.

1. **Multiply.** Build the 64-bit products $P_0 = M_0 x_0$ and $P_1 = M_1 x_2$, with the odd multipliers $M_0 = \text{D2511F53}_{16}$, $M_1 = \text{CD9E8D57}_{16}$.
2. **Split.** Cut each product into a high and a low 32-bit half, $P_j = \text{hi}_j 2^{32} + \text{lo}_j$: bits travel from the low counter words to all 64 positions.
3. **Fold.** Mix a high half, a passing lane and a subkey with xor: $y_0 = \text{hi}_1 \oplus x_1 \oplus k_0$ and $y_2 = \text{hi}_0 \oplus x_3 \oplus k_1$.
4. **Swap the lows** across the lanes: $y_1 = \text{lo}_1$ and $y_3 = \text{lo}_0$.

Why a bijection? M_0, M_1 odd $\Rightarrow$ invertible modulo 2^{32} : any y has a unique predecessor. Bijectivity avoids collisions and gives a clean period.

Philox: one round — wiring



$$(y_0, y_1, y_2, y_3) = (hi_1 \oplus x_1 \oplus k_0, lo_1, hi_0 \oplus x_3 \oplus k_1, lo_0),$$

where (hi_j, lo_j) are the high/low 32-bit halves of $M_j x_j$.

Philox: the key schedule (Weyl sequence)

Each round must use **different subkeys**, otherwise round symmetry would leak structure. Between two rounds, the subkeys advance by a fixed **Weyl step**:

$$(k_0, k_1) += (W_0, W_1)$$

with the constants

$$W_0 = 9\text{E}3779\text{B}9_{16}, \quad W_1 = \text{B}\text{B}67\text{A}\text{E}85_{16},$$

taken as odd parts of well-chosen irrationals (W_0 from the golden ratio φ , W_1 from $\sqrt{3} - 1$).

- A single 64-bit integer addition per round: **cheap**.
- The schedule is invertible, so each round stays a bijection.

Philox: from blocks to a number sequence

After $r = 10$ rounds, the counter/key pair has been fully scrambled into a block $R = E_K(C)$. We map a 32-bit word to $[0, 1)$ with a plain division:

$$u = \frac{R_j}{2^{32}} \in [0, 1),$$

so each call provides up to 4 uniforms (or a 128-bit block).

Summary of the [streaming model](#):

- draw next block: increment the counter, $C \leftarrow C + 1$;
- jump m blocks ahead: $C \leftarrow C + m$ (an integer addition);
- switch stream: change the key K ;
- no state to store or synchronize: natural for GPUs and parallel workers.

Counter-based generators: parallel streams

- The key identifies an independent stream, and the counter its position: jumping ahead amounts to integer addition.
- Stateless: no internal array to store or synchronize; the value at position i does not depend on the draw order. Well-suited to GPUs and parallel Monte Carlo.
- Reproducibility: identical (counter, key) yields identical bits on all architectures.
- Natural fit for stochastic optimization: one stream per scenario or sample index.

Adoption:

- NVIDIA cuRAND and PyTorch (GPU): Philox4 $\times$ 32–10;
- TensorFlow: Philox by default, Threefry optional;
- NumPy `Generator`: Philox bit generator; JAX: Threefry.
- Modern ML leans on it: dropout masks and distributed training draw thread- or rank-independent numbers.

Counter-based vs conventional generators

	Conventional (MRG, PCG, xorshift)	Counter-based (Philox, Threefry)
Stored state	full generator state	counter + key
Workload	transition function	output function (rounds)
Jump ahead	dedicated algorithms	addition on the counter
Independent streams	dedicated machinery	distinct keys
Draw order	sequential	position-indexed
GPUs	serial dependency	stateless threads

Jump ahead

- A very useful possibility proposed by some implementation is the possibility to make a jump of m positions in the random number sequences, with m very large.
- This allows one to easily define independent random variables.
- Useful in simulation when relying on common random numbers.

The MRG32k3a implementation proposes functions to generate independent streams and substreams.

RandomDataStreams.jl

<https://github.com/JLChartrand/RandomDataStreams.jl>

Documentation:

<https://jlchartrand.github.io/RandomDataStreams.jl/dev/>

This package proposes an implementation of MRG32k3a with streams and substreams, following the same philosophy as counter-based generators: explicitly indexed, reproducible subsequences rather than synchronized states.

Illustration (one independent stream per scenario, one substream per replication):

```
u = draw(stream = s, substream = r, position = i)
```

Non-uniform random variables generation

A good reference: Luc Devroye, *Non-Uniform Random Variate Generation*,
<http://luc.devroye.org/rnbookindex.html>.

Assume that we have a good uniform random variates generator, but we want to generate random variables following various probability laws: Normal, Weibull, Poisson, binomial,...

The desired properties are:

- correct method (or good approximation);
- as simple as possible, but as fast as possible;
- low memory consumption;
- robust;
- compatible with variance reduction technique (as quasi-Monte Carlo).

Inversion

Preferred method when applicable: compatible with variance reduction and copulae.

Consider a random variable X with cdf F . Let $U \sim U(0, 1)$ and

$$X = F^{-1}(U) = \min\{x : F(x) \geq U\} \quad (\text{Generalized inverse}).$$

Then

$$P[X \leq x] = P[F^{-1}(U) \leq x] = P[U \leq F(x)] = F(x),$$

i.e., X has the desired distribution. Indeed,

- in the continuous case, $F(X) \sim U[0, 1]$;
- in the discrete case, it is easy to prove that $\forall i, P[X = x_i] = p(x_i)$, and we assume $x_1 < x_2 < \dots < x_n$;
- The principle still works for mixed distributions.

Inversion (2)

- **Advantage**: monotone, only one U for all X .
- **Weakness**: for some laws, F is very difficult to invert. But we can often approximate F^{-1} .

Example: normal law.

If $Z \sim N(0, 1)$, then $X = \sigma Z + \mu : N(\mu, \sigma^2)$.

It is therefore sufficient to be able to generate a $N(0, 1)$, of density

$$f(x) = (2\pi)^{-1/2} e^{-x^2/2}.$$

We do not have any formula for $F(x)$ or $F^{-1}(x)$. Efficient codes however exist to approximate $F^{-1}(x)$.

Chi-square, gamma, beta, etc.: it is much more complicated since the form of F^{-1} depends of the distribution parameters.

Inversion for discrete distributions

Recall that

$$p(x_i) = P[X = x_i]; \quad F(x) = \sum_{x_i \leq x} p(x_i).$$

We have to generate U , search $I = \min\{i | F(x_i) \geq U\}$ and return x_I .

Various algorithms perform this search. Their efficiency depends of the distribution.

Initialization: store the x_i and $F(x_i)$ in arrays, for $i = 1, \dots, n$.

1. **Linear search** (time in $O(n)$): $U \leftarrow U(0, 1)$; $i \leftarrow 1$;
while $F(x_i) < U$ do $i \leftarrow i + 1$; return x_i .

2. **Binary search** (time in $O(\log(n))$):

$U \leftarrow U(0, 1)$; $L \leftarrow 0$; $R \leftarrow n$;

while $L < R - 1$

$m \leftarrow \lfloor (L + R)/2 \rfloor$;

if $F(x_m) < U$ then $L \leftarrow m$ otherwise $R \leftarrow m$;

return x_R .

Other approaches: composition

Assume that F is a convex combination of several cumulative distribution functions:

$$F(x) = \sum_{j=0}^{\infty} p_j F_j(x),$$

and that it is easier to invert F_j , $j = 0, \dots, \infty$ than F .

Generate $J = j$ with the probability p_j , then generate X following F_j .

The method therefore requires two uniforms for each random variable, and exploit the decomposition

$$P[X \leq x] = \sum_{j=1}^{\infty} P[X \leq x | J = j] P[J = j] = \sum_{j=1}^{\infty} F_j(x) p_j.$$

Convolution

Convolution. Assume that

$$X = Y_1 + Y_2 + \dots + Y_n,$$

where the Y_i are independent, of given laws. We generate the Y_i , $i = 1, \dots, n$, and we sum.

Examples: Erlang (sum of exponentials with same mean), binomial.

Acceptance/rejection: the most important technique after inversion.

We consider the case where X is continuous (the discrete case is analogous). Let $f(x)$ be the density of X , and let t be a "hat" function that majors f , i.e.

$$f(x) \leq t(x) \quad \forall x.$$

Acceptance/rejection

We can normalize t in a density r :

$$r(x) = t(x)/a, \text{ where } a = \int_{-\infty}^{\infty} t(s)ds.$$

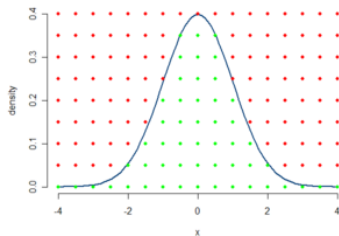
We choose t so that

1. it is easy to generate random variables of density r ;
2. a is small (close 1), or in other terms, $t(x)$ is close to $f(x)$.

The choice of t may be automatized.

Algorithm: Repeat

1. generate Y of density $r(x)$;
2. generate $U : U(0, 1)$ independently of Y ;
3. until $U \leq f(Y)/t(Y)$;
4. return Y .



Particular cases

Sometimes, we can benefit of mathematical transformations.

The main weakness is that they are seldom compatible with variance reduction techniques.

Example: Box-Muller method for the normal law.

Idea: it is easier to generate a point (X, Y) from the bivariate normal law, of density on $\mathbb{R}^2$

$$f(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}.$$

We change the cartesian coordinates (X, Y) by the polar coordinates (R, Θ) :

$$R^2 = X^2 + Y^2; Y = R \sin \Theta.$$

It gives an elegant approach, but incompatible with variance reduction techniques and can be slower than inversion.

Box-Muller Algorithm

1. Independently draw U_1, U_2 from a $U(0, 1)$ distribution.
2. Set

$$R = \sqrt{-2 \log(U_1)}$$

$$\theta = 2\pi U_2$$

3. Set

$$X = R \cos(\theta)$$

$$Y = R \sin(\theta)$$

Multivariate distributions

What if we want to draw from $\mathbf{X} : \Omega \rightarrow \mathcal{X} \subseteq \mathbb{R}^d$?

General approach:

- Generate the marginals
- Combine the marginals as desired.

Multivariate distributions: simple cases

- Ideally, we can explicitly transform the marginals.
- Example: multivariate normal $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_d \end{pmatrix} \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} & \dots & \sigma_{1,d} \\ \sigma_{1,2} & \sigma_2^2 & \dots & \sigma_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1,d} & \sigma_{2,d} & \dots & \sigma_d^2 \end{pmatrix}$$

Then,

$$\mathbf{X} = \boldsymbol{\mu} + L\mathcal{N}(\mathbf{0}, \mathbf{I})$$

with $\Sigma = LL^T$, and L is the Cholesky factor.

Drawing from $\mathcal{N}(\mathbf{0}, \mathbf{I})$: generate d $\mathcal{N}(0, 1)$ independently.

Multivariate distributions: copulae

A d -variate **copula** $C : [0, 1]^d \rightarrow [0, 1]$ is the cdf of a random vector $(U_1, \dots, U_d)$ with $U(0, 1)$ margins:

$$C(u) = P[U_1 \leq u_1, \dots, U_d \leq u_d]$$

where

$$P[U_j \leq u_j] = u_j$$

for $j = 1, \dots, d$, and $0 \leq u_j \leq 1$.

Sklar's theorem

Theorem (Sklar's theorem (1959))

- *Let H be a multivariate distribution function with margins $F_1, \dots, F_d$. There exists a copula C such that*

$$H(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)), \quad x_1, \dots, x_d \in \overline{\mathbb{R}}.$$

If F_i , $i = 1, \dots, d$, are continuous then C is unique. Otherwise C is uniquely determined on the Cartesian product of the range of the marginals $F_1(\overline{\mathbb{R}}) \times \dots \times F_d(\overline{\mathbb{R}})$.

- *Conversely, if C is a copula and $F_1, \dots, F_d$ are univariate distribution functions, then the function H defined above is a multivariate distribution function with margins $F_1, \dots, F_d$.*

Copula

If we know the joint CDF F and the marginals $F_1, \dots, F_d$, we can find the copula via

$$C(u_1, \dots, u_d) = F \left(F_1^{-1}(u_1), \dots, F_d^{-1}(u_d) \right),$$

where $F_j^{-1}(\cdot)$ is the generalized inverse for margin j :

$$F_j^{-1}(u) = \min\{x : F_j(x) \geq u\}.$$

Fréchet–Hoeffding bounds

Any bivariate copula C verifies

$$\max(F_X(x) + F_Y(y) - 1, 0) \leq C(x, y) \leq \min(F_X(x), F_Y(y))$$

Upper bound proof

$$\begin{aligned} C(x, y) &= P[X \leq x \cap Y \leq y] \\ &\leq \min(P[X \leq x], P[Y \leq y]) = \min(F_X(x), F_Y(y)) \end{aligned}$$

since $\{X \leq x\} \cap \{Y \leq y\} \subseteq \{X \leq x\}$ and $\subseteq \{Y \leq y\}$.

Fréchet–Hoeffding bounds

Lower bound proof

$$\begin{aligned}1 - C(x, y) &= P[X > x \cup Y > y] \\&\leq P[X > x] + P[Y > y] \\&= 1 - F_X(x) + 1 - F_Y(y)\end{aligned}$$

Thus,

$$C(x, y) \geq F_X(x) + F_Y(y) - 1.$$

Note: this can be viewed as a particular of the Bonferroni inequality

$$P\left[\bigcap_{i=1}^d A_i\right] \geq 1 - d + \sum_{i=1}^d P[A_i].$$

Gaussian copula

$$C(F_1(x_1), \dots, F_d(x_d)) = \Phi_{\Sigma} \left(\Phi^{-1}(F_1(x_1)), \dots, \Phi^{-1}(F_d(x_d)) \right),$$

where

- Φ is the cdf of a $\mathcal{N}(0, 1)$,
- Φ_{Σ} is the cdf of a d -dimensional $\mathcal{N}(\mathbf{0}, \Sigma)$.

Copulae estimation

- Parametric estimation;
- Non-parametric estimation;
- Machine-learning models (see Jutras et al.:
<https://arxiv.org/abs/2302.09193v3>).